

JOE KARAM

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Open to relocate internationally; will require sponsorship, but bring global engineering experience across the US and UAE that adds strong value.

EDUCATION

The Pennsylvania State University

Masters of Engineering in *Engineering Design (focus on Product Realization)*
3.9/4.0

Aug 2022

Pennsylvania, USA

California State University – Chico

Bachelor of Science in *Mechanical Engineering*
3.6/4.0 (junior & senior)

May 2021

California, USA

PROFESSIONAL EXPERIENCE

Creative Technologist

Sep 2023 – Present

Digitas (Publicis Groupe)

Dubai, UAE

- Owned the end-to-end execution of engineering and experiential projects, partnering with clients (Nissan, AHB...) to shape scope, schedules, budgets, and negotiations
- Consulted on the design and development of products for internal Publicis initiatives, creating concept sketches, CAD models, high-fidelity renders, and supporting prototyping, supplier sourcing, material selection, and manufacturability
- Designed and developed a limited-edition, fashion-forward car key for the 2025 Nissan Kicks, combining wearable aesthetics with automotive innovation
- Led the customization of an off-the-shelf arcade cabinet for Nissan, reengineering its internal hardware and integrating a custom retro game to reinforce brand engagement.
- Directed R&D efforts to embed advanced technologies, data analysis, and AI tools into digital platforms for clients like Bab Rizq Jameel and Nissan
- Created technology-enhanced, user-centered environments/products for ADCB and Al Hilal, architecting interactive spaces that blend cutting-edge tech with intuitive design

Product Design Consultant – Founder and Lead Designer

Aug 2024 – Present

Self-employed

Dubai, UAE

- Leading clients through the complete product development lifecycle, from initial brainstorming and sketching to final production, by understanding their vision and providing technical guidance throughout the process
- Creating detailed 3D models, and developing prototypes utilizing additive manufacturing to test form, fit, and function
- Researching and identifying appropriate suppliers, materials, and manufacturing methods to optimize product performance and cost-efficiency, ensuring a smooth transition from prototype to production

Hardware Design Engineer

Oct 2022 – Aug 2023

Cleo Robotics Inc.

Massachusetts, USA

- Created and optimized multiple prototype iterations to enhance the product's functionality and manufacturability
- Partnered with the US Department of Defense and improved the drone's capabilities in military surveying applications
- Utilized Solidworks to design and develop drone components (including chassis and sub-assemblies), leveraging 3D printing technologies (SLA and FDM) and carbon-fiber CNC machining
- Programmed and employed high-precision sensors to collect and analyze extensive data, enabling the optimization of CFD studies
- Collaborated seamlessly with cross-functional teams (programming/electronics), to ensure the feasibility and practicality of designs

Computational Design Intern

May 2022 – Aug 2022

Craftnetics Inc.

California, USA

- Spearheaded the development of a patent-pending computational tool for automated Additive Manufacturing, providing new solutions to manufacturing challenges
- Created a web-based application using CAD API, streamlining human-centered design processes

Mechanical Engineering Research Assistant

Mar 2020 – Dec 2020

California State University - Chico

California, USA

- Formulated a 3D convex hull algorithm in Grasshopper (visual programming CAD software)
- Modeled CAD geometries in Rhinoceros 3D

PROJECTS

The Learning Factory's Vending Machine (PSU Learning Factory)

Aug 2021 – May 2022

- Led a dynamic software/electrical team of four individuals in delivering solutions for controlling applications
- Employed user experience methodologies (interviews, user-testing) to align product features with specific objectives
- Delivered a reliable, user-friendly, and engaging product while successfully meeting the client's requirements

Design for Additive Manufacturing (DfAM)

Aug 2021 – Dec 2021

- Created a Design for Additive Manufacturing (DfAM) framework, assisting beginners in successfully 3D printing models & wrote a research paper about it
- Explored and integrated opportunistic and restrictive thinking to design for 3D printing, encompassing both plastics and metals

Compliant and Intelligent Grasping with Parallel Kinematic Mechanism

Aug 2020 – May 2021

- Modeled the system's chassis and conducted both static and dynamic Finite Element Analyses
- Implemented detailed design for manufacturing and successfully built the robot's chassis
- Managed and documented project expenses, and developed the Bill of Materials (BOM) & 2D CAD drawings

EXTRACURRICULAR ACTIVITIES

Penn State Robotics Club, Team Member and Recruiter

Aug 2021 – May 2022

- Supported the team of educational robotics, instructing motor control fundamentals
- Offered my CAD and 3D printing skills to design and produce parts for various applications

CSU Chico Baja Racing Club, Team Member and Educator

Aug 2020 – May 2021

- Contributed as a cage designer for the University's offroad automotive racing club
- Assisted the club by teaching around 10 newcomers CAD techniques in Rhinoceros 3D

CORE TECHNICAL SKILLS

Languages: English (Fluent), French (Fluent), Arabic (Native)

CAD Software: SolidWorks, Rhinoceros 3D (+ Grasshopper), Autodesk Fusion 360, Blender ...

Languages (Programming): Python, R (statistics), C++ (Arduino), MATLAB, HTML, CSS, JavaScript

Extra Skills: Human-Centered Design, Parametric Design, UAV (assembly, troubleshooting, testing), Additive (FDM, SLA) and Traditional (CNC, Milling...) Manufacturing, Rendering, FEA (static and dynamic), Data Manipulation, Drafting, R&D (innovation), MS Office (ppt, excel...)